

PAPER DEMONSTRATION

SYNAPSE-S2

Visual Operator Manual

A guided tour of the Namespace Galaxy, cortex, ganglia, neurons, governed recall, memory maintenance, and reliability controls.

THE CORE IDEA

The Galaxy is a read-only neural atlas. Use it to locate and understand memory. Use Cortex Governor, Memory Write, Recall, Memory Graph, and governed bridge controls to change durable state.

READ-ONLY ATLAS

GOVERNED WRITES

EXPLICIT PRUNE

VERIFIED RECOVERY

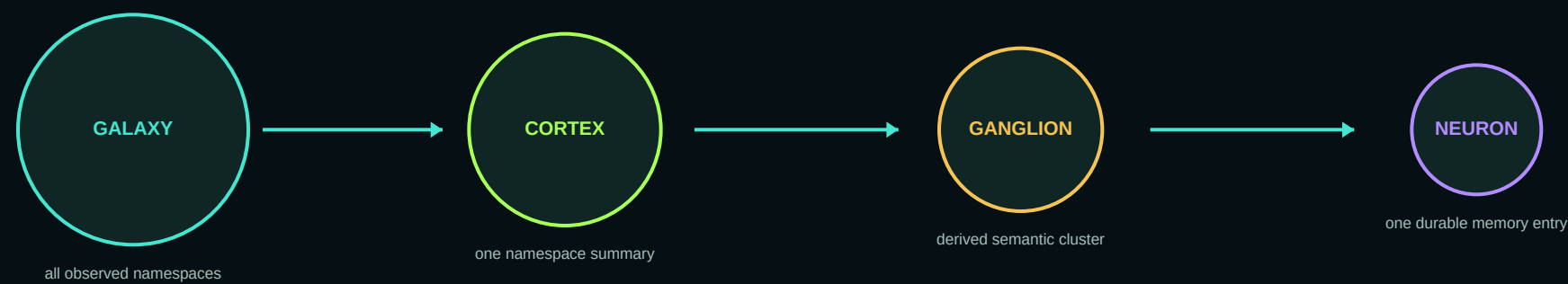
Interface shown: release 018cd32db4013dee7763f1517f953eb956b7fa0d

Live dashboard examples supplied July 29, 2026

Budget proposal supporting document

The memory anatomy

A visual level of detail is not a mutation boundary. Each inward step reveals more structure while preserving the same durable namespace.



VISUAL MASS FORMULAS

Galaxy

58% memory volume
27% indexed density
15% approved bridges

Cortex

72% bounded log memories
28% relationships per memory

Ganglion

68% relative log memories
32% weighted edge density

Neuron

Relative log visible weighted degree
inside the returned sample

WHAT SIZE MEANS

- A relative visual comparison of stored volume and connectivity.
- A fast way to spot dense or highly connected areas.
- A navigational cue that changes as the selected scope changes.

WHAT SIZE NEVER MEANS

- Truth, confidence, importance, quality, recency, or permission.
- A command to retain, delete, promote, or bridge memory.
- Proof that every relationship in the database is currently visible.

All namespaces: the Galaxy

READ-ONLY + SCOPE

CONNECTED MEMORY
Namespace Galaxy
Every saved context remains isolated until an approved bridge deliberately connects it.

14 namespaces 1 bridges 11 pending / suggested

SELECTION
default
This is the active sidebar namespace. Local recall stays here, with only explicitly global memory inherited.

Memories	3,534
Indexed surface terms	162,938
Indexed term density	46.11 terms / memory
Active bridges	0
Incident enabled bridge weight	0.00
Relationships	3,618
Relationship density	1.02 / memory
Relative size score	78%

Size formula: 58% relative log memory · 27% relative log indexed density · 15% relative log bridges

Events: 5,056
Last update: Jul 28, 2026, 5:06 PM

Enter namespace Current sidebar context

Evidence-only suggestions
Review first; suggestions never expand recall automatically.

IT-OPS-WORKLOG	related_to: 0.46
PTZPLZ	related_to: 0.45
demo	related_to: 0.44
servus-servus-gui-sanitized-handoff-28268638	related_to: 0.41

Legend:
 ● Sphere area: weighted memory mass
 — Bridge width: approved weight
 — Higher approved weight
 --- Disabled bridge
 --- Pending proposal (isolated)
 --- Suggestions never affect size or recall

Galaxy size: 58% relative log memory volume, 27% relative log indexed term/relationship density, 15% relative log enabled approved bridge weight centrality. Density blends 55% indexed surface terms/memory with 45% relationships/memory when both exist, missing inputs are omitted and remaining weights normalize. Internal layers state their formula and sample scope in the inspector.

► Browse all namespaces as a list

1 Namespace spheres

Sphere area combines memory volume, indexed density, and enabled approved bridge centrality.

2 Bridge and suggestion lines

Only approved enabled bridges grant Connected recall. Suggestions stay isolated.

3 Selection inspector

Read totals, density, formula, update time, and bridge evidence before acting.

4 Active sidebar context

This scope governs later capture, recall, graph, and Cortex actions.

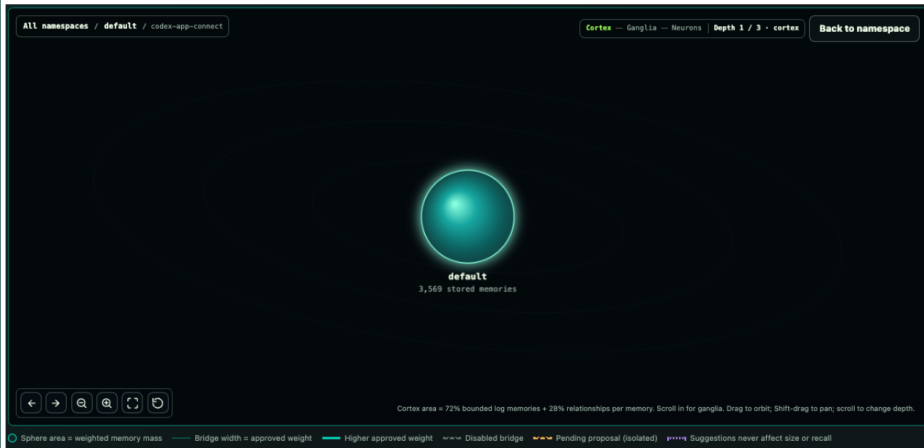
NAVIGATION

- Drag orbit; Shift-drag pan.
- Scroll or +/- changes depth.
- Enter/Space opens or focuses.
- Escape/Back moves outward.
- F fits; R resets.

Inside one namespace

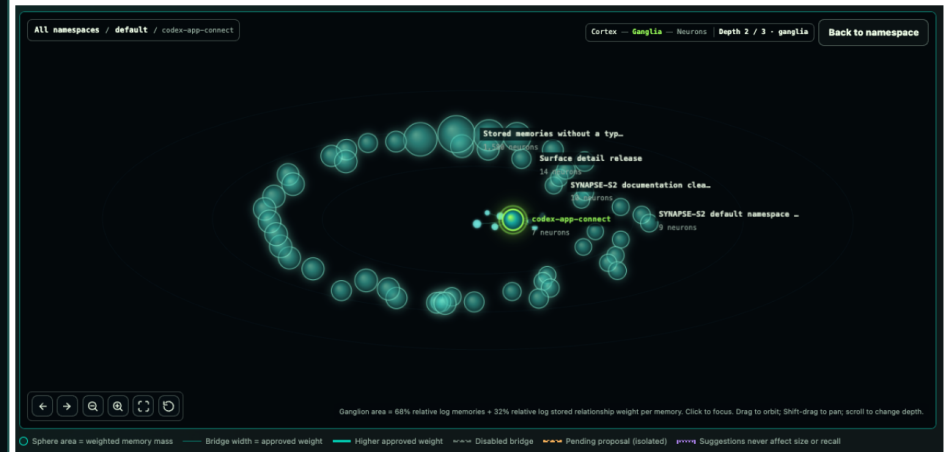
Scroll inward to change semantic level of detail. The namespace remains the same; only the projection changes.

“Cortex view of my “default” namespace.”



DEPTH 1 / 3 - CORTEX

“Ganglia view”



DEPTH 2 / 3 - GANGLIA

CORTEX VIEW

- One sphere summarizes the selected namespace.
- Area = 72% bounded log memories + 28% relationships per memory.
- Use this layer to confirm the context boundary before working.
- Scrolling inward reveals ganglia; nothing is rewritten.

GANGLIA VIEW

- Each sphere is a derived semantic/type cluster.
- Area = 68% relative log memories + 32% weighted edge density.
- Clusters recompute from stored metadata and relationships.
- There is no direct create, rename, merge, reassign, or delete ganglion action.

What “Focus this ganglion” does

READ-ONLY

SELECTION

Stored memories without a typed namespace

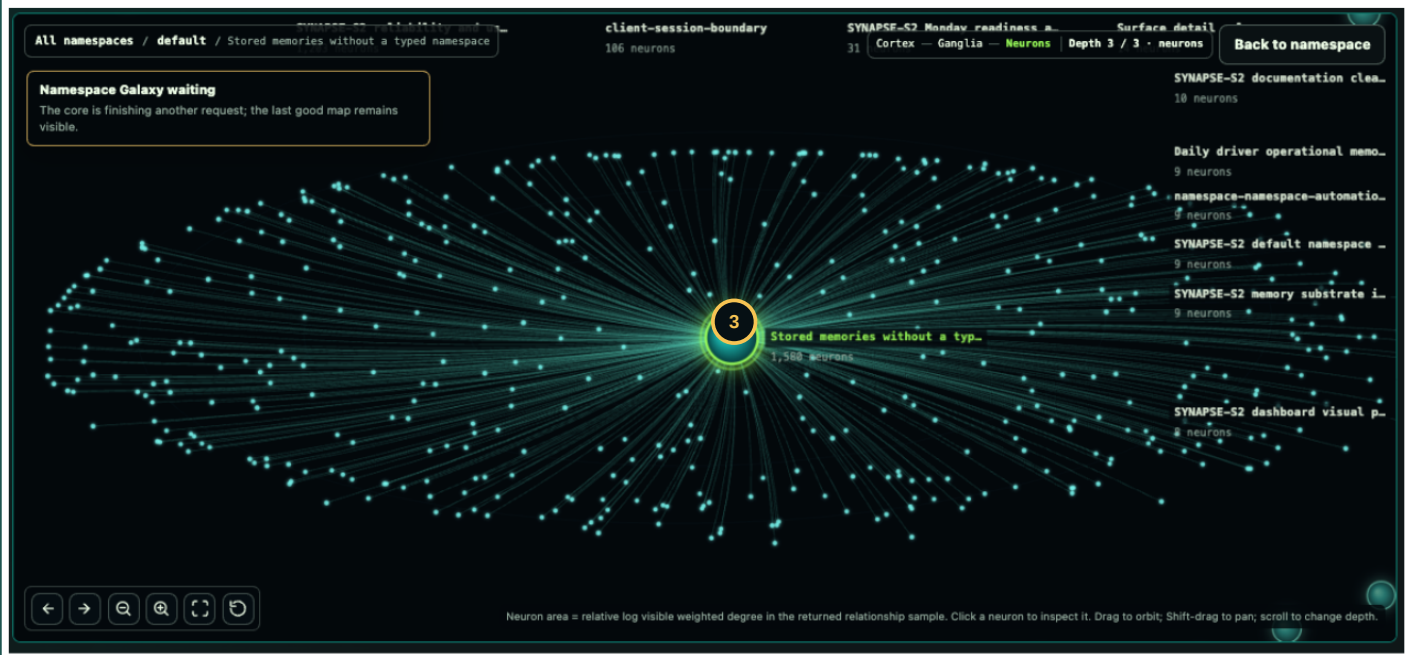
Stored semantic cluster. Area combines its reported memory total with server aggregate inter-ganglion weight; the inspector names lower-bound or visible-sample scope.

1

Neurons (memories)	1,580
Memory total scope	Stored total
Inter-ganglion relationships	0
Aggregate edge weight	0.00
Weighted relationship density	0.00 / memory
Relationship metric scope	stored aggregate
Relative size score	68%
Size formula	68% relative log memories · 32% relative log weighted density
Cluster basis	fallback_untyped
Stored types	memory: 1580
Last updated	Jul 29, 2026, 1:31 PM

Focus this ganglion

2



FOCUS CHANGES

Selection, camera, zoom, bounded detail request, breadcrumbs, and browser history. It centers the ganglion, resets pan, raises zoom to neuron depth, and fetches a bounded node/edge sample.

FOCUS DOES NOT CHANGE

Memory, relationships, confidence, recall scope, bridges, acknowledgement, or retention. Outside-cluster nodes may remain dimmed only for visual orientation.

1 Derived cluster detail

The inspector reports stored membership, density, basis, types, and last update.

2 The Focus action

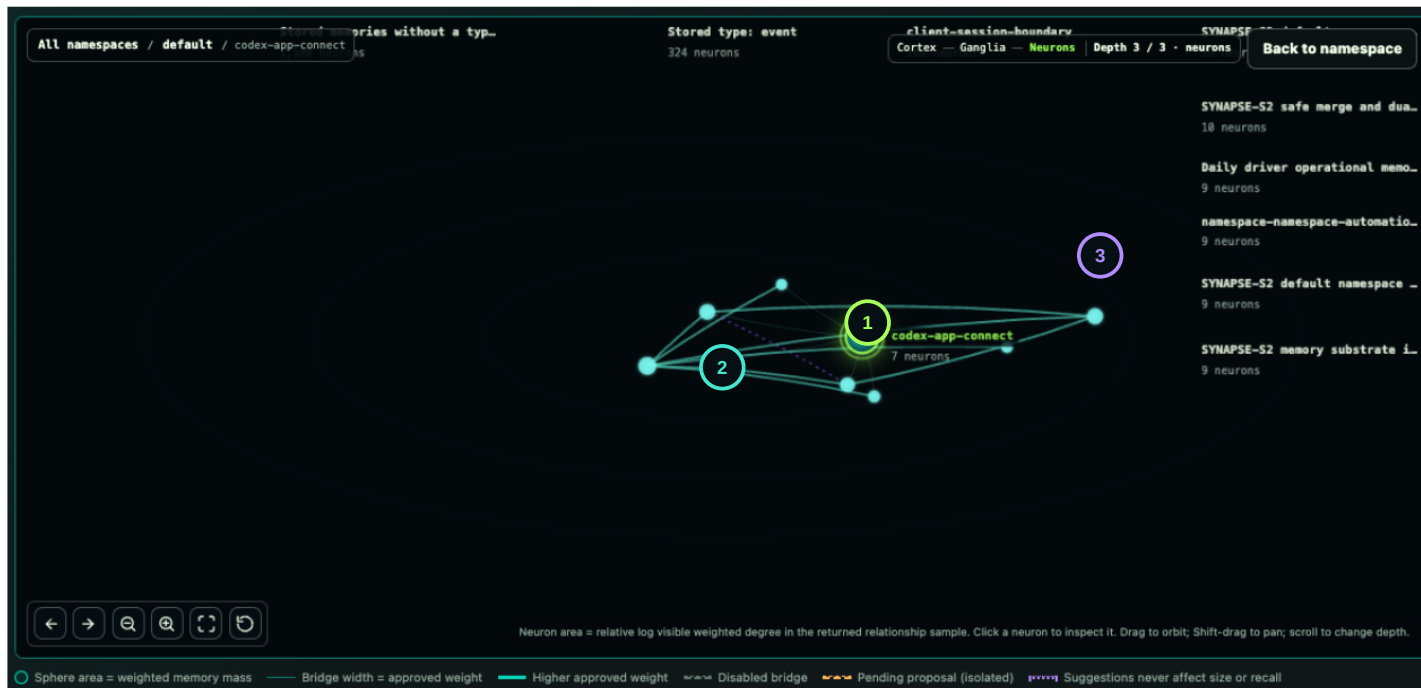
Loads a bounded neuron projection. It is a navigation command, not a memory command.

3 Focused center

The chosen ganglion becomes the center of the neuron projection.

Neuron view: inspect durable memory

"Neuron view"



1 Selected neuron

One durable memory entry. The inspector can show type, tag, source, time, excerpt, and returned relationships.

2 Visible edge sample

Lines show relationships returned for this bounded projection. Visible degree is not guaranteed to be global degree.

3 Relative node size

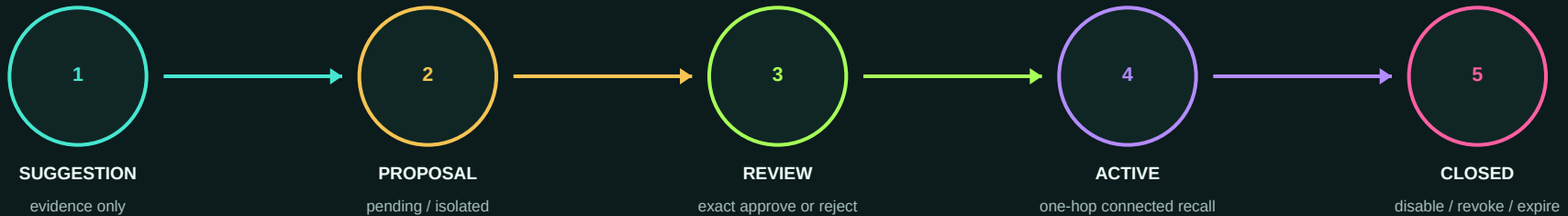
Neuron area reflects relative log visible weighted degree inside the returned sample.

TO CHANGE A NEURON

- Promote/demote only a governed Cortex trace.
- Capture a corrected replacement when facts changed.
- Prune the exact bad node or edge in Memory Graph.
- Create recovery evidence before broad deletion.

Connect namespaces with governed bridges

BRIDGE LIFECYCLE



Only stage 4 grants recall authority. Suggestions and proposals never expand recall. Rejected, revoked, and expired states preserve audit history.

RECALL CONSOLE

Deterministic hybrid retrieval

[Clear](#)

What this does

Embeds your text locally, reads durable spike and semantic indexes, fuses versioned rank signals, and returns bounded matching traces without mutating memory state.

Where it goes

Local uses this namespace plus global memory; Connected adds approved one-hop bridges; All searches every namespace.

Good prompts

Ask for prior decisions, incidents, people, project state, or temporal context you have already captured.

RECALL SCOPE

☒ **Local**
Current + global memory

☐ **Connected**
Approved one-hop bridges

☐ **All**
Every namespace

Local searches the active namespace plus explicitly global memory. This is the safe default.

K= 8

[▶ Recall](#)

Results (0)

Context: active · Latency: -- ms

RECALL SCOPES

- **Local:** active context plus explicit global memory. Safe default.
- **Connected:** Local plus approved enabled one-hop bridges.
- **All:** every namespace for one explicit read-only search.

HARD GUARANTEES

- No cross-namespace memory copy or synchronization.
- No wildcard, multi-hop, or automatic approval.
- Bridge width/opacity visualizes approved weight only.
- Changing weight or direction requires a new reviewed proposal.

Capture and retrieve deliberately

MEMORY WRITE

Durable trace capture

5,097 published context updates

Ready for Remember/Ingest handoffs via leased-at-least-once; 12 acknowledged / 0 leased

TRACE TAG

trace-tag

SOURCE TAG

source-tag

SESSION TAG

codex-session

SPEAKER

codex

TRACE TEXT

Live context text

SURPRISE

0.58

SENTENCES

1

CONVERSATION NOTES

Paste real user/Codex decisions, corrections, temporal notes, or session summaries

EVENT TEXT

Sequential briefing text

Remember + publish

Ingest + publish

Capture conversation

OPERATION LOG

Backend response stream

Backend snapshot

{
 "context_id": "default",
 "runtime": "ready",
 "effective_enabled": true,
 "memory_entries": 3567,
 "relationships": 3640,
 "context_bus_events": 5097,
 "memory_mb": 288.09375,
 "server_total_ms": 1611.508,
 "client_hydrate_ms": 1623.7,
 "generated_at": 1785353505.2498481
}

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RECALL SCOPE

Local

Current + global memory

Connected

Approved one-hop bridges

All

Every namespace

Local searches the active namespace plus explicitly global memory. This is the safe default.

Ask memory: decision, incident, project state...

K= 8

Recall

Results {0}

Context: active · Latency: -- ms

READ-ONLY RETRIEVAL

CREATION PATHS

- Remember + publish: one durable trace plus context update.
- Ingest + publish: structured events and relationships.
- Capture conversation: exactly-once bounded capture.
- App Connect: preview local UI text, then explicitly snapshot or capture a selection.

RETRIEVAL CONTRACT

Retrieval v2 is deterministic and read-only. It does not run recurrent updates, STDP, pruning, activity mutation, or cache mutation.

Always verify the active sidebar context before writing. That context determines where durable memory is stored.

SYNAPSE-S2 Visual Operator Manual | Local first; expand to Connected or All only when needed

08

Cortex Governor: govern the work

GOVERNED WRITE

AGENT GOVERNANCE
Cortex Governor

cognitive_governance:strict

No Session

Cortex is idle, not broken

No agent is currently governed. Enter the current task before mutating files, capturing risky data, or making handoff claims.

Start Work

Enter Cortex

Tick before risky actions

Commit / Wrap / End

SESSION STATE

Not started

Start Cortex Session before risky work

LAST DECISION

standby

tick records proceed / verify / sanitize

TYPED MEMORY

follow_up:213

goal / decision / validation / risk

GUARDRAILS

clear

mutation and sensitive-data warnings

NEXT MOVE

Enter Cortex Governor ... tied to a session.

governed recommendation

ENTER CORTX

Put the agent under governed memory

AGENT

dashboard-ui

MODE

strict

CURRENT TASK

Describe the work this agent is about to perform

Start Cortex Session

GOVERNOR TICK

Check the next move against memory

OBSERVATION

What did the agent observe?

PROPOSED ACTION

What is the agent about to do?

INTENDED FILES

One path or glob per line

INTENDED TOOLS

One command or tool per line

CONFIDENCE

0.62

Mutation intent

Tick governor

COMMIT CORTICAL TRACE

Store verified thoughts as typed memory

TYPE

evidence

TRUTH POSTURE

observed

TRACE TEXT

Verified decision, constraint, validation result, or risk to preserve

Commit trace

High-confidence truths	Assumptions and conflicts	Capture queue	Working memory
No high-confidence governed truths yet	No unresolved assumptions or conflicts	No pending capture recommendations	client-session-boundary-ev... follow_up / observed / 0.76 conf... SYNAPSE-S2 MCP client... Promote Demote Prune cortex-codem-desktop-falls... follow_up / observed / 0.76 conf... SYNAPSE-S2 MCP client... Promote Demote Prune client-session-boundary-ev... follow_up / observed / 0.76 conf... SYNAPSE-S2 MCP client... Promote Demote Prune cortex-codem-desktop-falls... follow_up / observed / 0.76 conf... SYNAPSE-S2 MCP client... Promote Demote Prune client-session-boundary-ev... follow_up / observed / 0.76 conf... SYNAPSE-S2 MCP client... Promote Demote Prune client-session-boundary-ev... follow_up / observed / 0.76 conf... SYNAPSE-S2 MCP client... Promote Demote Prune cortex-codem-desktop-falls... follow_up / observed / 0.76 conf... SYNAPSE-S2 MCP client... Promote Demote Prune cortex-codem-desktop-falls... follow_up / observed / 0.76 conf... SYNAPSE-S2 MCP client... Promote Demote Prune

FOUR-STEP LIFECYCLE

- 1

Start

agent, mode, current task
- 2

Tick

observation, action, files/tools, mutation intent
- 3

Commit

typed trace plus truth posture
- 4

Close

end session or wrap a handoff

MODERATION

Promote raises governed confidence; Demote marks stale; Prune deletes the governed trace after confirmation.

Manage memory at the exact blast radius

MEMORY GRAPH

Event relationships

Nodes **3,567**

Edges **3,640**

Temporal **42**

Associative **38**

Events **104**

Last **19m ago**

Prune

Neural Inspector

42 temporal / 38 associative

GRAPH PRUNE

Remove bad memory graph data

Event/node memory-id relationship-id event-id tag fallback reason **Confirm prune**

Clear temporal edges **Clear associative edges**

RELATIONSHIP EDGES

temporal_next Agent: codex-desktop -> Context: default client-session-boundary / agent / client-session-boundary / context	1.000	Jul 23, 02:48 PM	Delete edge
temporal_next Context: default -> Session ID: 699cd8c44ee24ee1b233227c093824ea client-session-boundary / context / client-session-boundary / session	1.000	Jul 23, 02:48 PM	Delete edge
temporal_next Session ID: 699cd8c44ee24ee1b233227c093824ea -> Reason: signal-sigterm client-session-boundary / session / client-session-boundary / reason	1.000	Jul 23, 02:48 PM	Delete edge

SUPPORTED PRUNE TARGETS

- One memory or event node.
- One exact relationship edge.
- One context / deployment event.
- All temporal edges in the context.
- All associative edges in the context.

Dragging a graph node changes only its screen position. It does not persist or rewire relationships.

SMALLEST CHANGE FIRST

Ganglia cannot be deleted directly because they are derived. Correct or prune exact member neurons and edges; the cluster recomputes.

SAFE DESTRUCTIVE CHECKLIST

1 Confirm context

2 Inspect ID + provenance

3 Create paired recovery

4 Prefer one node/edge

5 Give a reason

6 Confirm once

7 Doctor + recall

Reliability is a set of independent gates

DIAGNOSTIC + FILESYSTEM

RUNTIME READY

SYNAPSE-S2 Core

CORE ENABLED

Spiking Engine

ACTIVE

Attention Router

ACTIVE

Memory Subsystem

ACTIVE

Dashboard API

LISTENING

DEVICE

MLX gpu

256 top-k / 8,192 neurons

CONTEXT MEMORY

3,567 traces

3,640 relationships / 12 contexts

GRAPH MODE

42 temporal / 38 associative

104 event traces visible

MAINTENANCE

standby

19m ago / 300s interval

MLX ready / mlxsnr ready / native LIF

Last tick: 01:31:45 PM

Hydrate: 1,611.5 ms api / 3,104 ms ui

CORE

RESOURCE ENVELOPE

Topology memory envelope

Refresh

Current

288.1 MB

96 MB

Minimum

384 MB

Maximum

OPTIMIZED OPERATING RANGE

Current

288.1 MB

inside 96-384 MB

Headroom

95.9 MB

384 MB ceiling

Trace Cache

4,527

registered

Arrays

5

tracked

SUBSTRATE

SHAPE / COUNT

MEMORY FOOTPRINT

W_lateral

8192 x 8192

256.000 MB

W_syn

1024 x 8192

32.000 MB

active_traces

8192

0.031 MB

mem

8192

0.031 MB

epk

8192

0.031 MB

QUICK PRUNE

standby

benchmark available

MAINTENANCE CONTROLS

Local actions

CORE CONTROL

Enable / Disable Core

Only control that changes runtime state

Unlock

Disable

Locked. Press Unlock before enabling or disabling SYNAPSE-S2 Core.

RELIABILITY CHECKS

Monday Readiness

Score demo-critical runtime paths

Score

Refresh Runtime State

Reload metrics and health

Refresh

Self Test

Check page, memory, capture, apps

Test

Readiness Audit

Probe runtime, graph, recall

Audit

Native Certify

Verify MLX, envelope, prune budget

Certify

Evidence Pack

Write report plus verified paired recovery bundle

Pack

Self test idle

Run a feature check before relying on the local control plane.

LOCAL INTAKE

App Connect

Detect apps and capture exposed UI text

Open

Magic Capture

Process local session inbox drops

Process

Capture inbox armed

Processed 0.000 files; last run captured 1 events from 1 payloads.

MAINTENANCE

DO NOT COLLAPSE THESE INTO ONE GREEN LIGHT

Runtime	Integrity	Delivery	Capture	Recovery
Core status and API availability.	SQLite, foreign keys, graph, and embeddings.	Lease, render, and exact receipt acknowledgement.	Inbox, exactly-once ledger, and unresolved debt.	Paired database/capture evidence and restore proof.

Quick Prune and Deep Sleep change transient neural state; neither directly deletes durable SQLite memories.

Find your way around the operator surface

OPERATOR ORIENTATION

Workbench

Operator Loop

Runtime

Memory Graph

Namespace Galaxy

Memory Write

Recall Console

Cortex Governor

App Connect

Maintenance

Logs

MEMORY CONTEXT

Choose a saved namespace or type a new one.

default (3,567)

SAVED NAMESPACES

default

Current memory URI: s2://local/default

Start Wizard

Recipes

App Connect

View Logs

DAILY OPERATOR LOOP

1

Choose context

→

2

Start Work

→

3

Enter Cortex

→

4

Tick risky action

→

5

Capture / Recall

→

6

Inspect / correct

→

7

Wrap + protect

GUIDED HELP

Start Wizard teaches the core workflow in real time. Recipes provides repeatable operations. App Connect captures exposed local UI text. Logs provides runtime evidence.

KEYBOARD AND POINTER CONTROLS

Drag
orbit

Shift-drag
pan

Scroll / +/-
zoom

Arrows
select

Enter / Space
enter or focus

Esc / Back
move outward

F
fit

R
reset

OPERATOR FLOW

1 Start Work

2 Enter Cortex

3 Tick Action

4 Capture / Recall

5 Wrap + Pack

HEALTH GOOD

MEMORY 288.1 MB / 384 MB

COMPUTE MLX gpu

CONTEXTS 12

TIME 01:31:45 PM

What operators can do - and where

Intent	Surface	Effect	Class
Browse / enter	Galaxy	Select and change active scope	Read / scope
Focus ganglion	Galaxy detail	Load bounded neuron projection	Read-only
Inspect neuron	Neuron inspector	Show provenance, excerpt, visible edges	Read-only
Recall	Local / Connected / All	Deterministic retrieval	Read-only
Store / ingest	Memory Write	Create durable traces and relationships	Write
Govern work	Cortex Governor	Start, tick, commit, close	Governed
Adjust trace	Cortex card	Promote / demote governed trace	Governed
Delete exact item	Memory Graph	Prune node, edge, or event	Destructive
Connect contexts	Bridge review	Propose and approve one-hop recall	Governed
Protect state	Recovery / Evidence	Paired verified artifacts	Filesystem

CURRENT BOUNDARIES

- No direct ganglion CRUD or neuron reassignment.
- No in-place stored-text editor or topology/weight canvas.
- No namespace rename, archive, or delete lifecycle.
- Bridges are namespace-level, not neuron or ganglion-level.
- No in-place approved bridge editing.
- Dashboard cannot disable/revoke a bridge or show full history.
- Promote/Demote applies only to governed Cortex traces.
- Quick Prune does not delete durable neurons.

These boundaries keep the atlas explainable and non-mutating. Durable changes pass through explicit, auditable surfaces.

FIVE-MINUTE PAPER DEMO

1 Galaxy

Choose a context

2 Cortex

Confirm the boundary

3 Ganglion

Focus a cluster

4 Neuron

Inspect one memory

5 Recall

Compare recall scopes

6 Governor

Show controlled mutation

7 Recovery

Close with evidence